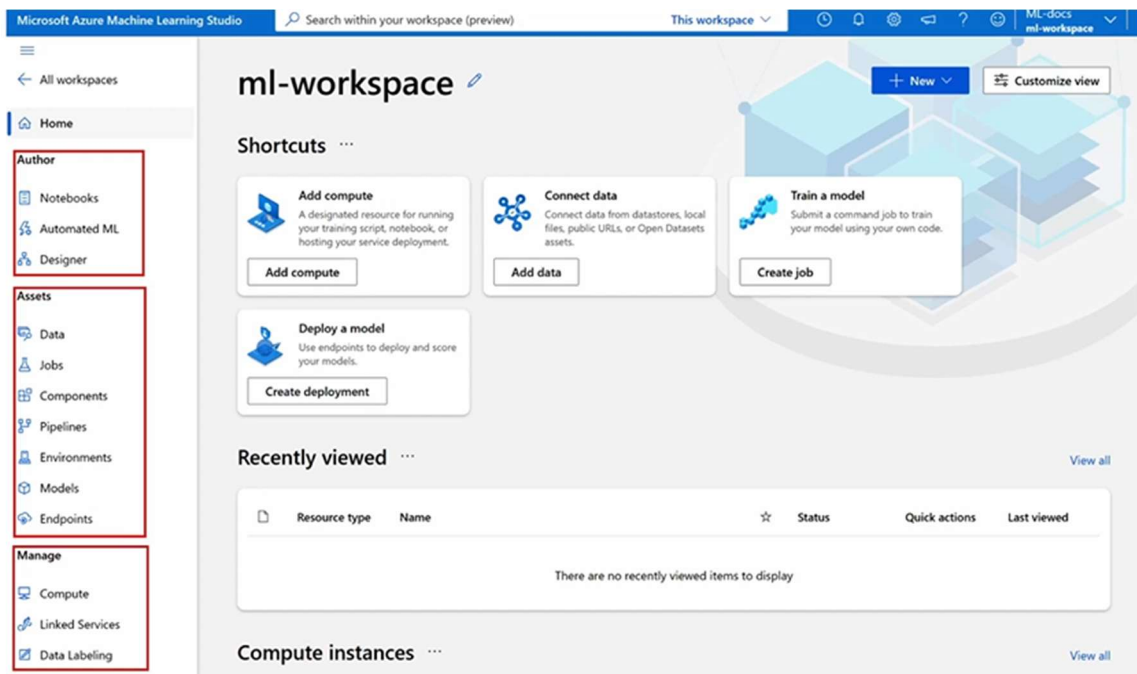


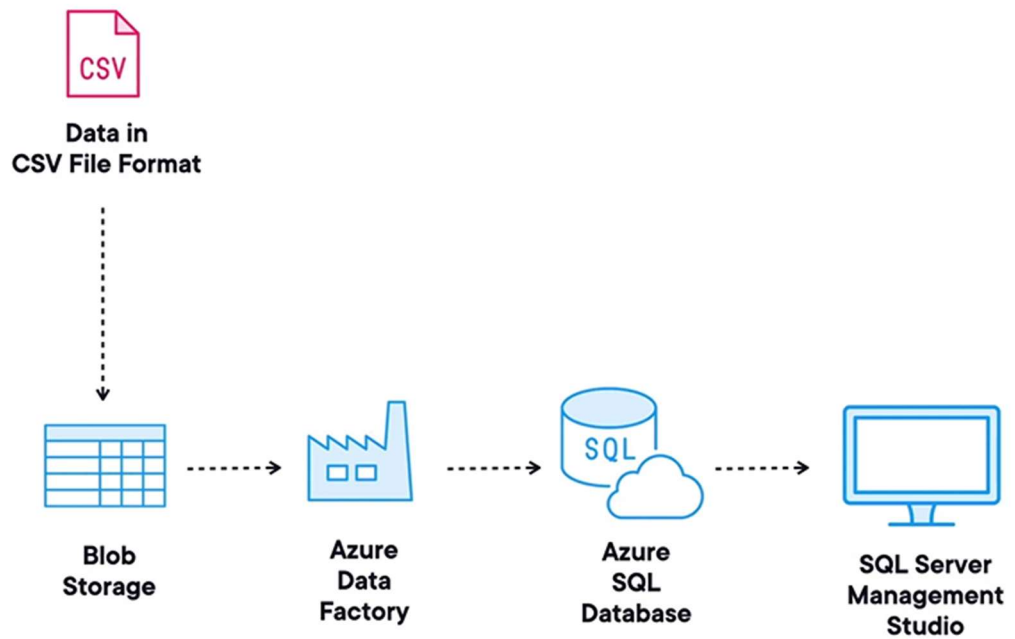
Discovering Azure machine Learning:

How to efficiently use Azure Ai for deploying ml

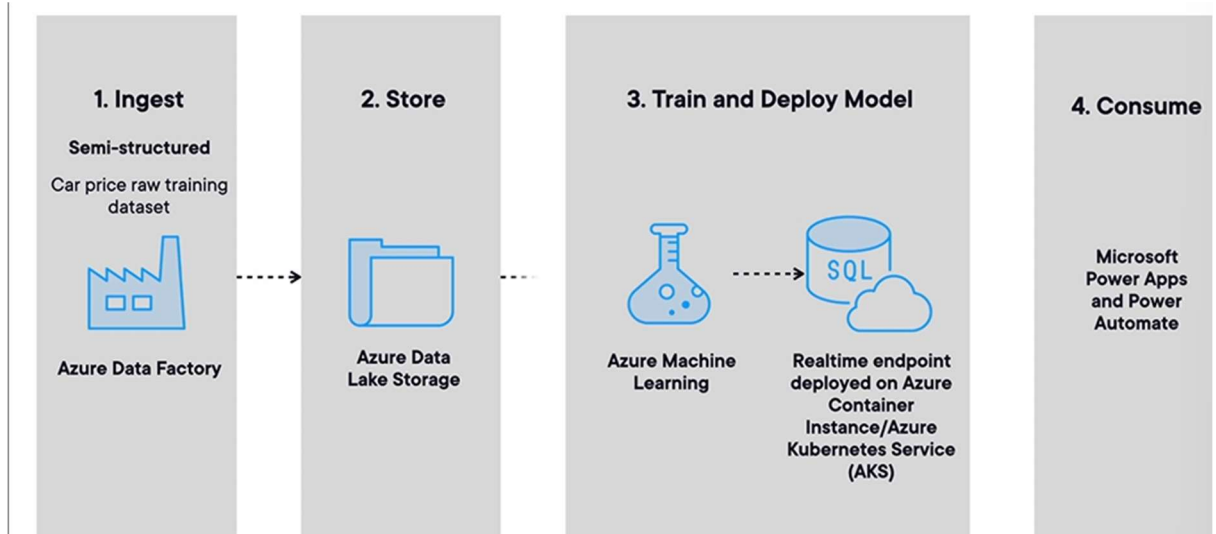
- ➔ Azure Machine learning:
- ➔ It is a comprehensive cloud based services
- ➔ Designed streamlined end to end machine learning lifecycle
- ➔ Integrated seamlessly other azure services allowing to build train deploy ml within robust azure system
- ➔ Azure ML workspace process and manages all resources :
- ➔ ML workspace is a headquarters where asset are stored and computed



- ➔ Azure ML offers a variety of computes offers : CPU ,GPU and Clusters
- ➔ Data stored in Azure ML functions as bridges connecting data stores with workspace



→ It makes sure the data is accessible to training and modelling



→ Powerful aspect of Azure ML as it integrates seamlessly with your Project

Building and Deploying Models:

➔ Model Catalog : is rich repository of previous models covering various ml task :

1ST

Classification

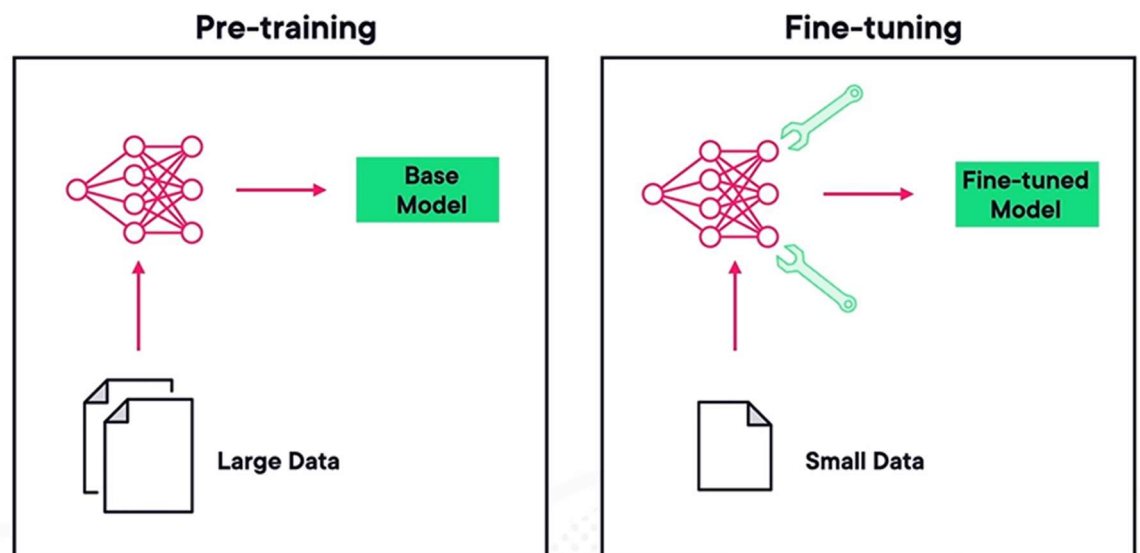
2ND

Regression

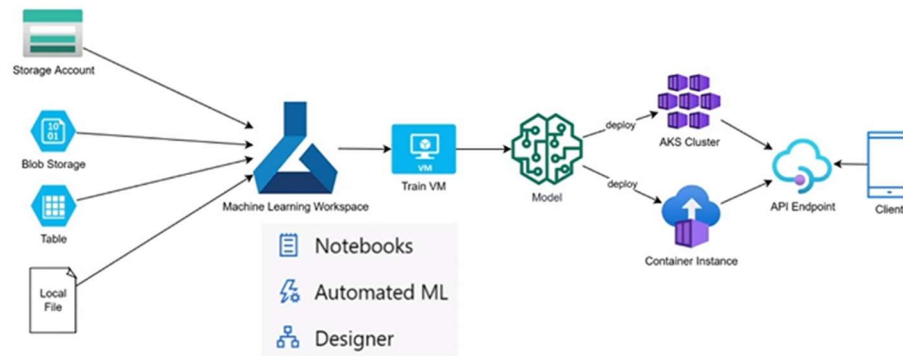
3RD

Clustering

➔ These can be utilized and covered directly within your code without writing from scratch



➔ You can directly fine tune it and use it with your training data



You can deploy your trained model as an endpoint, which can be used for real-time

Building , Deploying and assessing a Model in Azure ML:

➔ Fine Tuning LLM and deploying

Steps:

- ➔ Open Azure ML
- ➔ Create a new resource

Microsoft Azure

Search resources, services, and docs (G+)

Home > Azure Machine Learning >

Azure Machine Learning

Create a machine learning workspace

Basics Networking Encryption Identity Tags Review + create

Resource details

Every workspace must be assigned to an Azure subscription, which is where billing happens. You use resource groups like folders to organize and manage resources, including the workspace you're about to create. [Learn more about Azure resource groups](#)

Subscription *

Resource group * [Create new](#)

Workspace details

Configure your basic workspace settings like its storage connection, authentication, container, and more. [Learn more](#)

Name *

Region *

Storage account * [Create new](#)

Key vault * [Create new](#)

Application insights * [Create new](#)

Container registry [Create new](#)

[Review + create](#) [< Previous](#) [Next : Networking](#)

- ➔
- ➔ Validating
- ➔ Launch Azure ML studio

Machine Learning Studio

Welcome to your new AzureML workspace! Quickly access recent and favorite resources, use quick actions, and customize the homepage to

azureml132425

+ New Customize view

Generative AI with Prompt flow

View prompt flow

Multi-Round Q&A on Your Data

Create a chatbot that uses LLM and data from your own indexed files to ground multi-round question and answering capabilities in enterprise chat scenarios.

[Start](#) [Clone](#)

Q&A on Your Data

Use LLM and data from your own indexed files to ground multi-round question and answering capabilities.

[Start](#) [Clone](#)

Web Classification

Use LLM to classify URLs into multiple categories.

[Start](#) [Clone](#)

Chat with Wikij

Create a chatbot that to ground the respo

[Start](#)

Generative AI models

View all

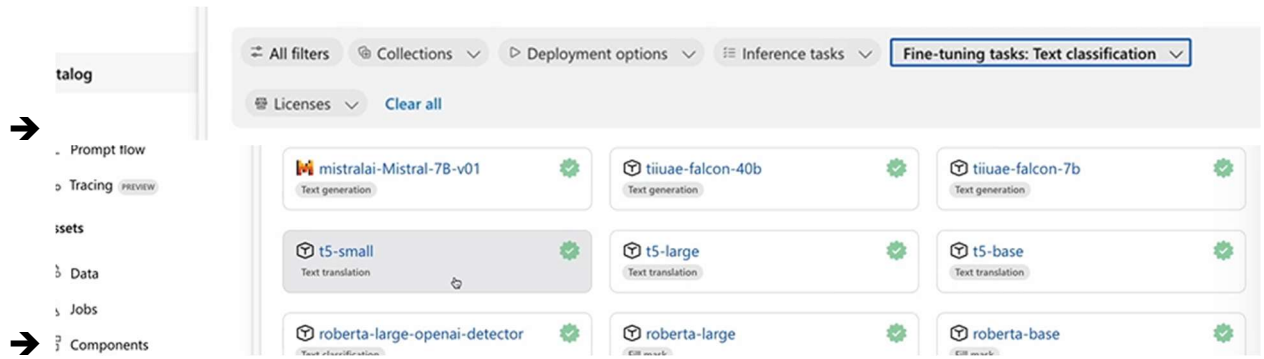
Phi-3-mini-4k-instruct

Phi-3-mini-128k-instruct

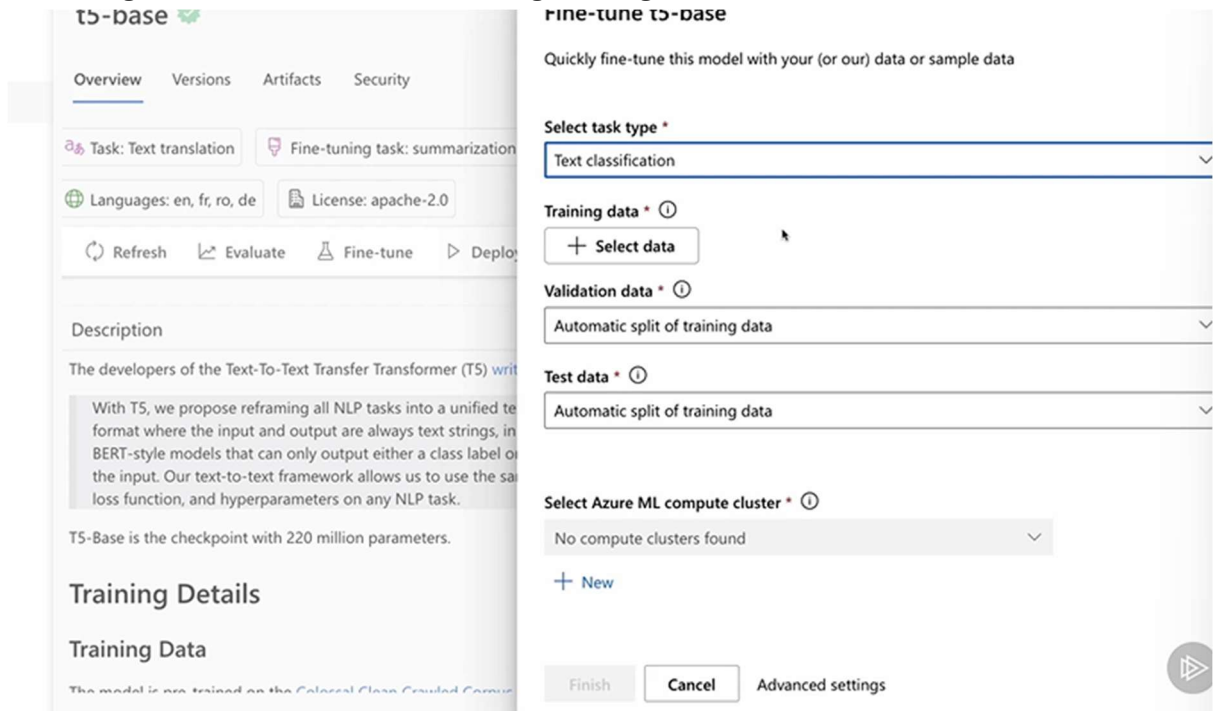
Llama-2-7b-chat

Phi-3-small

- ➔ Now we fine tune llm that does sentiment analysis



→ Selecting T5 base that loads fine tuning catalog



- Upload your csv file
- Training data

Select training data

Select data

Map data

Map the columns in your data

business_id	date	review_id	stars	text	type	user_id	cool	useful	funny	Column11	Column12
9yKzy9PA...	2011-01-26	fWKvX83p...	5	"My wife t...	null	null	null	null	null	null	null
Do yoursel...	null	null	null	null	null	null	null	null	null	null	null
While EVE...	I had the ...	null	null	null	null	null	null	null	null	null	null

Text classification data needs to contain at least 2 columns: one for 'Sentence1' (string) and another for 'Label' (integer / string).

Required data columns *

Select the 'sentence1 key' column *

Select or search by name

Select the 'label key' column *

Select or search by name

Optional data columns

Select the 'sentence2 key' column

Select or search by name

Back

Add

Cancel



then adding dataset

Fine-tune t5-base

Quickly fine-tune this model with your (or our) data or sample data

Select task type *

Text classification

Training data *

yelp_csv_2024-07-13_201530_UTC

User data

Sentence1Key: text

LabelKey: stars

Validation data *

Automatic split of training data

Test data *

Automatic split of training data

Select Azure ML compute cluster *

No compute clusters found

+ New



➔ Adding cluster and can add nodes in the cluster

Create compute cluster ⓘ

Advanced Settings

Location *
East US

Virtual machine tier ⓘ
☐ Dedicated ☒ Low priority

Virtual machine type ⓘ
☐ CPU ☒ GPU

Virtual machine size ⓘ
☒ Select from recommended options ☐ Select from all options

Name ↑	GPU device	Workload types	Available quota ⓘ	Cost ⓘ
☐ Standard_NC6s_v3 6 cores, 112GB RAM, 736GB storage	1 x NVIDIA Tesla V100 16GB vRAM	Single-node testing or training of deep learning models	150 cores	\$0.61/hr
☒ Standard_NC12s_v3 12 cores, 224GB RAM, 1474GB storage	2 x NVIDIA Tesla V100 16GB vRAM	Single-node, high-scale training of deep learning models	150 cores	\$1.22/hr
☐ Standard_NC24s_v3 24 cores, 448GB RAM, 2948GB storage	4 x NVIDIA Tesla V100 16GB vRAM	Multi-node testing or training of deep learning models	150 cores	\$2.45/hr
☐ Standard_ND40rs_v2 40 cores, 672GB RAM, 2900GB storage	8 x NVIDIA Tesla V100 32GB vRAM	Multi-node, high-scale training of deep learning models	150 cores	\$4.41/hr

Back Next

Create compute cluster ⓘ

Virtual Machine
Advanced Settings

Configure Settings
Configure compute cluster settings for your selected virtual machine size.

Name	GPUs	Cores	Available quota	RAM	Storage	Cost/Node
Standard_NC12s_v3	2 x NVIDIA Tesla V100	12	150 cores	224 GB	1474 GB	\$1.22/hr

Compute name * ⓘ
azuremlcompute132425

Minimum number of nodes * ⓘ
0

Maximum number of nodes * ⓘ
1

Idle seconds before scale down * ⓘ
30

☐ Enable SSH access ⓘ

> Advanced settings

Add tags ⓘ
Name : Value Add
☐ No tags

Back Create Download a template for automation. Cancel

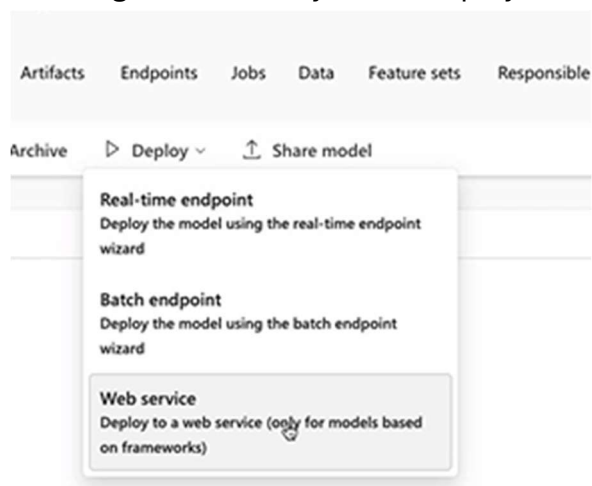
➔ This process takes more than 1 hour to complete the training

➔ The register the models

Register model from a job output

The screenshot shows a four-step wizard for registering a model from a job output. The steps are: 1. Select job (completed), 2. Select output (current step), 3. Model settings, and 4. Review. In the 'Select output' step, a green message box states: 'A named model output has been detected in the job outputs and auto-selected. You can select any other output and/or type if this is not what you intend to select.' Below this, the 'Model type' dropdown is set to 'MLflow'. The 'Job output' field contains the text 'trained_model (azureml_6c8d0c77-86f3-4132-86c0-bff16c2b4f48_output_mlf...' with a refresh icon to its right.

➔ Then register and the you can deploy



➔

➔ Web services – when using web framework

➔ Batch – endpoint – when using azure services

→ Select instance of virtual machine

Current Project resource
discoverazureml

Virtual machine * ⓘ

Standard_DS3_v2 4 Cores, 14 GB (RAM), 28 GB (Disk), \$0.29/hr

Showing 61 VM sizes

Name ↑	Category	Av...	Cost ⓘ
☐ Standard_DS4_v2 8 cores, 28GB RAM, 56GB storage	General pur...	50 co...	\$0.58/...
☐ Standard_F2s_v2 2 cores, 4GB RAM, 16GB storage	Compute o...	50 co...	\$0.08/...
☐ Standard_F4s_v2 4 cores, 8GB RAM, 32GB storage	Compute o...	50 co...	\$0.17/...
☐ Standard_F8s_v2 8 cores, 16GB RAM, 64GB storage	Compute o...	50 co...	\$0.34/...
☐ Standard_NC12s_v3	GPU	50 co...	\$6.12/...

Inferencing data collection ⓘ

☒ Disabled

Package Model ⓘ

☒ Disabled

Deploy Cancel More options

→ Set instance count -3

→ Then deploy and the process takes 10 min

```
{
  "0": {
    "label": 1,
    "score": 0.09899949216842651
  },
  "1": {
    "label": 2,
    "score": 0.130886048078537
  },
  "2": {
    "label": 3,
```

→

Review the original model card to understand the inputs, outputs, data used to train the model, evaluation metrics, license, intended uses, limitations, and risks. [View model](#)

Sample inference

Input *

```
{
  "input_data": [
    "This is the worst restaurant!"
  ],
  "params": {
    "return_all_scores": true
  }
}
```

Test

VIEW

